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Global Coreference Resolution via Local Resolution and Quotient-Graph Inference

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Abstract

This paper demonstrates that global entity resolution does not require full-document quadratic self-attention mechanisms. Instead, we show that the coreference relation can be factorized into local window-based clustering and transitivity-driven inference over an induced quotient graph. Furthermore, we challenge the paradigm of monolithic universal language models for coreference by decomposing the required signal space into independent, frozen specialist encoders whose vector outputs act as a basis for a learned composition head. Our two-stage pipeline performs quadratic attention strictly within small, localized windows (Stage A) and projects the resulting local clusters into a quotient space for cross-window matching (Stage B). With all underlying encoders completely frozen, this architecture achieves a definitive CoNLL F_1 score of 0.8556 on the CoNLL-2012 benchmark using gold mentions, while providing substantial computational and memory improvements over traditional full-document architectures.

Keywords: coreference resolution, representation factorization, quotient graph, frozen language models, window-based inference, transitive closure, efficient inference

1 Introduction

Coreference resolution aims to recover an equivalence relation over a set of nominal mentions $\mathcal{M} = \{m_1, \dots, m_n\}$ in a document, where $m_i \sim m_j$ indicates that two mentions refer to the same underlying entity. The induced partition $\mathcal{E} = \{E_1, \dots, E_k\}$ defines the entity structure of the document.

Modern state-of-the-art systems execute coreference resolution by running full-document self-attention over thousands of subtokens jointly. This incurs an $O(N^2)$ computational cost, driven by the assumption that a monolithic universal model must process the entire context to disentangle overlapping semantic and contextual signals.

This paper demonstrates that this quadratic overhead is unnecessary, guiding the design via two core hypotheses. First, Coreference exhibits local edge locality but global transitive structure. The overwhelming majority of informational dependencies are satisfied within short text spans, and long-range connections can be mathematically reconstructed via transitive closure. Consequently, quadratic attention is only required within small, localized windows. Second, a monolithic model is not required to observe the entire document if the necessary signal dimensions are pre-separated into independent channels. By utilizing frozen, specialized models whose representations function like an informational basis across a vector space, a lightweight linear head can resolve entities without global cross-attention.

The remainder of this paper is organized as follows. Section 2 outlines the overall workflow of the two-stage pipeline. Section 3 develops the mathematical foundations of representation factorization and locality-transitivity equivalence. Sections 5 and 6 details the technical implementation of the Stage A local clusters and Stage B quotient-graph matching modules. Section 7 describes our multi-corpus training environment, and Section 8 presents our empirical results alongside comprehensive diagnostic proofs of our core hypotheses.

2 Solution Overview

To bypass global quadratic text attention entirely, the framework processes an incoming document through a decoupled, two-stage hierarchy that transforms a dense mention pool into a series of sparse, cluster-level operations over a quotient graph.

1. **Decoupled Encoding Stream:** The document's nominal mentions are mapped once to raw semantic vectors (via a frozen text encoder) and localized context vectors (via a frozen language model). These representations are kept distinct and never fused inside a monolithic layer.

2. **Stage A (Intra-Window Clustering):** Suppose a document of length N is partitioned into w windows of lengths $l_1, \dots, l_w$ satisfying

$$\sum_{i=1}^{w} l_i = N.$$

The total self-attention cost is

$$\sum_{i=1}^{w} O(l_i^2).$$

By Cauchy-Schwarz,

$$\left(\sum_{i=1}^{w} l_i\right)^2 \leq w \sum_{i=1}^{w} l_i^2.$$

Therefore

$$\sum_{i=1}^{w} l_i^2 \geq \frac{N^2}{w}.$$

Equality holds iff

$$l_1 = l_2 = \cdots = l_w = \frac{N}{w}.$$

Thus equal-sized windows minimize the total quadratic attention cost among all partitions with w windows. Full quadratic self-attention and pairwise mention scoring run exclusively within these local boundaries. A local union-find execution connects matching mentions, collapsing each window's sub-chains into a set of discrete, local entity clusters.

3. **Quotient Graph Transformation:** The system lifts the problem space from individual mentions to the cluster level. The newly resolved local clusters are treated as atomic nodes in a highly compressed quotient graph.

4. **Stage B (Cross-Window Quotient Matching):** Cross-window entity consistency is recovered by evaluating edges across the quotient nodes. Rather than re-comparing individual mentions across long distances, a size-invariant interaction head evaluates cluster-level compatibility augmented with an additive, independent lexical channel. Global transitive closure over these accepted cluster edges outputs the final document-level entity partition. The point here is that while quadratic reasoning is still there, it is transferred onto a compressed state space which requires a heavily reduced number of computation rendering the effective complexity feasible.

3 Theoretical Foundation and Design Principles

3.1 The Representation Factorization Hypothesis

Monolithic language models incur extreme computational costs because they are forced to end-to-end ingest whole-document sequences to implicitly separate intersecting semantic, contextual, and discourse dependencies. We argue that this universal parameter updates can be completely bypassed if the broader relational task can be structurally transformed into localized sub-problems mapped to a global quotient graph. Under this factorization, a universal model is no longer required to learn overlapping dimensions on its own. Instead, we propose a more general informational framing:

Hypothesis 1 (Low-Redundancy Information Basis Hypothesis) *Let the complete decision space $\mathcal{V}$ required to unify local relational partitions into a global equivalence graph be spanned by d distinct feature channels. If a global task can be* *decomposed into independent local window assignments and aggregated via a macro quotient space, the information space can be factorized into discrete, separate channels:* $\mathcal{V} = \mathcal{V}_1 \times \mathcal{V}_2 \times \cdots \times \mathcal{V}_d$. *Crucially, these channels must remain as close to linearly independent as possible.*

The channels need not be mathematically orthogonal. Instead, each channel must contribute information that cannot be reconstructed from the others. Empirically, introducing channels that are largely recoverable from existing channels provides little or no improvement and may introduce optimization noise. The objective is therefore not orthogonality but maximal marginal information gain. A diagnostic experiment illustrates this effect. For mention pairs with different surface lemmas, the raw cosine similarity of the underlying axis vectors achieves only AUC □ 0.57. Using the same vectors as independent inputs to a learned linear probe achieves AUC = 0.91. The signal is therefore already present in the basis coordinates. Collapsing them to a scalar similarity destroys most of the discriminative information.

3.2 Locality-Transitivity Equivalence

The necessity for full-document quadratic attention rests on the assumption that long-range mention interactions must be explicitly modeled at the token level. We replace this with a local graph formulation:

Hypothesis 2 (Transitive Link Reconstruction) *Let* $\sim$ *be the global coreference equivalence relation over a document. For any two distant mentions* $u, v \in \mathcal{M}$ *belonging to the same entity, there exists a finite chain of intermediate references* $m_1, m_2, \dots, m_r$ *such that* $u \sim m_1 \sim \cdots \sim m_r \sim v$*, where each adjacent pair in the chain is co-located within a local window partition:* $\pi(m_j) = \pi(m_{j+1})$.

Theorem 1 (Quotient Reconstruction) *Assume Hypothesis 2 holds and that Stage B correctly identifies all cross-window cluster correspondences. Then the connected components of the quotient graph coincide exactly with the global entity partition* $\mathcal{E}$.

Proof Stage A defines an equivalence relation $\sim_A$ restricted to local windows W_k. If u and v are coreferent and reside in different windows, Hypothesis 2 guarantees a path of local links connects them. Stage B computes cross-window mapping edges between local equivalence classes. Since the union of these local relations and cross-window matches forms an equivalence relation, its connected components over the quotient nodes coincide exactly with the global entity partition $\mathcal{E}$. ■

This structural choice is supported by the geometric decay of similarity bounds over long token spans. For three mention vectors where $\cos(\mathbf{u}, \mathbf{v}) \geq c$ and $\cos(\mathbf{v}, \mathbf{w}) \geq c$, the strict mathematical lower bound on the endpoints degrades to:

$$\cos(\mathbf{u}, \mathbf{w}) \geq 2c^2 - 1$$

Under this constraint, two adjacent links at $c = 0.80$ bound the endpoints to a minimum similarity of just 0.28. Direct mention comparison across distant windows quickly approaches a zero-signal floor. Transitive closure over discrete quotient components (union-find) is therefore required to propagate entity chains without signal decay.

4 Model Class Definition

We formalize the system as a structured inference model over a partitioned mention space. Let a document be represented as a sequence of tokens from which a set of mentions $\mathcal{M} = \{m_1, m_2, \dots, m_N\}$ is extracted. The document is partitioned into a collection of disjoint, non-overlapping windows $\mathcal{W} = \{W_1, W_2, \dots, W_K\}$ such that $W_i \cap W_j = \varnothing \ \forall i \neq j$ and $\bigcup_{i=1}^{K} W_i = \mathcal{M}$. Each mention belongs to exactly one window via a deterministic assignment function $\pi : \mathcal{M} \to \mathcal{W}$.

Let $\mathbf{h}_i = \left[f_{\theta_1}(m_i) \parallel f_{\theta_2}(m_i) \right]$ denote a factorized mention representation produced by independent frozen specialist encoders. Within each window W_k, a local coreference relation $\sim_k \subseteq W_k \times W_k$ is computed via a localized scoring function s_θ. Two mentions are linked if $m_i \sim_k m_j$ if and only if $s_\theta(\mathbf{h}_i, \mathbf{h}_j) > \tau$. Let $\sim_k^*$ denote the transitive closure of this local relation, inducing a partition $\mathcal{C}_k = W_k / \sim_k^*$. Stage A produces a collection of disjoint local clusters $\mathcal{C} = \bigcup_{k=1}^{K} \mathcal{C}_k$. We define a quotient mapping $q : \mathcal{M} \to \mathcal{C}$ inducing the quotient space $\mathcal{Q} = \mathcal{C}$. Each cluster $C \in \mathcal{Q}$ is treated as an atomic node in subsequent inference, restricting all cross-window reasoning strictly to the quotient space $\mathcal{Q}$ rather than the raw mention pool $\mathcal{M}$.

5 Stage A: Local Intra-Window Resolution

The document is partitioned into fixed non-overlapping windows of size $K = 256$ subtokens. Neither encoder backbone is fine-tuned; RoBERTa-large (355M parameters) and BGE-large (335M parameters) remain completely frozen. For each mention i, the contextual channel supplies the RoBERTa token vectors at the span start $\mathbf{c}_i^{(s)}$ and end $\mathbf{c}_i^{(e)}$, while $\mathbf{s}_i$ denotes the BGE vector of the surface text. Each channel is projected independently:

$$g_i = [\, P_c([\mathbf{c}_i^{(s)} \mid \mathbf{c}_i^{(e)}]) \mid P_s(\mathbf{s}_i) \,] \in \mathbb{R}^{2048}$$

where $P_c \in \mathbb{R}^{1024 \times 2048}$ and $P_s \in \mathbb{R}^{1024 \times 1024}$ are learned linear layers. Pairwise link scoring within a local window is modeled via a multi-layer perceptron:

$$s(i, j) = \mathrm{MLP}_p \left([\, g_i \mid g_j \mid \mathbf{d}_{ij} \,] \right)$$

where $\mathbf{d}_{ij} \in \mathbb{R}^{32}$ is a learned distance embedding. Training optimizes a Mention-ranking Marginal Log-Likelihood (MLL) over the window partitions, where each mention i softmaxes over $\{\varepsilon\} \cup \{j < i\}$, with ε serving as a learned null antecedent:

$$\mathcal{L}_i = -\log \sum_{j \in \mathrm{Gold}(i)} \frac{\exp(s(i,j))}{\sum_{k \in \{\varepsilon\} \cup \{j<i\}} \exp(s(i,k))}$$

Transitive closure is applied via union-find within the window boundaries to generate the initial quotient cluster space $\mathcal{C}$.

6 Stage B: Cross-Window Quotient Matching

Stage B reconciles entity fragments across window pairs, operating entirely over the quotient graph nodes produced by Stage A. It scores cluster pairs directly over their full cross-cluster mention grid via an architecture optimized for fine-grained interaction (`MentionMatcher`).

6.1 Size-Invariant Log-Mean-Exp Interactivity

Collapsing clusters into a single vector via attention pooling can introduce an informational smearing effect, where highly specific local evidence—such as a shared unique proper noun—is averaged out across a cluster's pronoun embeddings. To preserve localized mention-pair evidence while preventing large clusters from artificially inflating connection scores, the `MentionMatcher` aggregates the full interaction grid via a size-invariant log-mean-exp operator:

$$\mathrm{score}(A,B) = \log \sum_{m \in A, n \in B} \exp\left(f(\mathbf{m},\mathbf{n})\right) - \log(|A| \cdot |B|)$$

where $f(\mathbf{m},\mathbf{n}) = \mathrm{MLP}([\mathbf{m} \mid \mathbf{n} \mid \mathbf{m} \odot \mathbf{n} \mid |\mathbf{m} - \mathbf{n}|])$, processing the member channel streams projected during Stage A. Subtracting $\log(|A| \cdot |B|)$ penalizes cluster-size scaling artifacts, ensuring that a single high-confidence mention-pair activation can safely anchor the cross-window bridge without being suppressed by cluster scale.

6.2 Lexical Features

To assist the frozen contextual spaces across deep token spans, the interaction head is augmented with an additive combination of localized, discrete string-identity features:

$$\sigma(A,B) = \mathrm{score}_{\mathrm{neural}}(A,B) + \mathbf{w}^\top \mathbf{x}_{\mathrm{lex}}$$

where $\mathbf{w}$ is a learned weight vector, and $\mathbf{x}_{\mathrm{lex}}$ contains:

1. **IDF-weighted token Jaccard similarity:** $\sum_{t \in A \cap B} \mathrm{idf}(t) / \sum_{t \in A \cup B} \mathrm{idf}(t)$, where document-local $\mathrm{idf}(t) = \log(M/(1 + \mathrm{df}(t))) + 1$.

2. **Substring containment:** A binary indicator marking if any content mention (excluding bare pronouns) in A is a substring of any content mention in B, or vice versa.

3. **Exact mention match:** A binary flag tracking if the two clusters share an identical content mention surface text string.

Final document clusters are constructed via global union-find over all cluster pairs whose final score $\sigma(A, B)$ exceeds a learned null bias parameter.

7 Experimental Setup

All evaluation experiments are conducted on the test split of the CoNLL-2012 English coreference benchmark (Pradhan et al., 2012) utilizing gold mention boundaries. Following empirical data mixture optimizations, our lightweight scoring modules are trained on a uniform corpus mixture consisting of CoNLL-2012, LitBank, CorefUD, and a subsampled partition of 8,000 documents from PreCo. Training enforces a uniform loss across all datasets without negative window-pair subsampling to preserve distributional balance. All baseline results cited from prior literature correspond strictly to their respective gold-mention evaluation configurations.

8 Results

8.1 Main Coreference Benchmarks

Table 1 establishes the primary system performance evaluated against state-of-the-art architectures using identical gold mention contexts.

Approach	**CoNLL**	**MUC**	**B^3**	**CEAFe**
Stage A Only (No Merge)	0.7439	0.8460	0.6910	0.6950
Stage A + B (Mention-Level)	**0.8556**	**0.9230**	**0.8310**	**0.8140**
Dobrovolskii (2021)	0.8180	0.8570	0.8000	0.7970
Caciularu et al. (2023)	0.8360	0.8740	0.8200	0.8130
Bohnet et al. (2023)	0.8380	0.8760	0.8220	0.8150
Luo et al. (2025)	0.8510	0.8890	**0.8350**	**0.8290**

Table 1: Full-document CoNLL-2012 test results on **gold mention boundaries**. Baseline systems are mapped strictly to their gold-mention settings to isolate linking capability.

Table 2 itemizes performance across granular cluster configurations, tracking the architectural stability of the model over varied mention types.

Cluster Category Type	CoNLL	MUC	B^3	CEAFe
PROPN-only	0.8390	0.8400	0.8380	0.8390
PRON+NOUN	0.7700	0.7630	0.7490	0.7970
NOUN-only	0.7500	0.7530	0.7500	0.7480
PRON+PROPN	0.7280	0.7450	0.6940	0.7430
all-mixed	0.7160	0.7730	0.6730	0.7030
NOUN+PROPN	0.7130	0.7190	0.7040	0.7150
PRON-only	0.6640	0.7050	0.6490	0.6370

Table 2: Definitive per-cluster category breakdown for the Stage A + B pipeline, CoNLL-2012 test.

8.2 Complexity Analysis

Since window size is fixed, $W = N/K$. Assuming mentions are approximately uniformly distributed, $m_w \approx M/W = MK/N$. Then total encoding complexity is

$$O\left(W * K^2\right) = O(NK)$$

instead of $O(N^2)$. in stage A, computation cost is

$$O\left(w\left(\frac{M}{w}\right)^2\right) = O\left(\frac{M^2K}{N}\right)$$

This is essentially linear until stage A. For stage B, complexity is

$$O(C^2)$$

Total complexity becomes

$$O\left(NK + \frac{M^2K}{N} + C^2\right)$$

For a typical document, we realistically have $C \ll M \ll N$. Therefore, even though quadratic computation is not fully eliminated, we only do it in a much more reduced space. Mean document statistics on CoNLL-2012: $N = 617.4$ tokens, $M = 57.1$ mentions $C = 20.7$ quotient nodes. The quotient graph contains only $20.7/57.1 = 36.3\%$ of the original mention nodes. Relative to token-level reasoning, the reduction is $617.4/20.7 = 29.8$ times per dimension, corresponding to roughly $(617.4/20.7)^2 \approx 890$ times fewer pairwise comparisons. The empirical results validate this generalized theoretical framework and information routing properties:

1. **Validation of Independent Basis Projections (Hypothesis 1):** Our core framework reaching a definitive 0.8556 F_1 score demonstrates that deep

Space	Nodes	Pairwise Comparisons
Token Space	617.4	381,183
Mention Space	57.1	3,260
Quotient Space	20.7	428

Table 3: Average document graph sizes on CoNLL-2012.

contextual models do not need to fine-tune internal weights to parse cross-cutting signals. By treating specialized universal embeddings as structural axes and preserving separate vector paths, the linear combination layer tracks identity without collapsing coordinates. This is strongly substantiated by our linear probe diagnostic, where keeping distinct vector coordinates yields an AUC of 0.91, compared to the near-chance level of ≈ 0.57 encountered when collapsing dimensions to scalar cosine steps.

2. **Validation of Locality (Hypothesis 2 & Proposition 1):** Stage A alone (operating strictly inside isolated windows without a global merging layer) yields a baseline score of 0.7439 F_1. This means that local-only tracking recovers approximately 87% (0.7439/0.8556) of the complete global framework performance. This explicitly confirms that entity dependencies are highly concentrated within bounded token ranges, proving that local quadratic attention is sufficient for core resolution.

3. **Validation of Quotient Graph Efficiency:** Evaluating the structure of the document space shows that Stage A collapses an average of 57.1 mentions per document into just 20.7 cluster nodes. This produces a structural compression profile of **2.76:1** before cross-window matching starts.

4. **Validation of Transitive Linking:** Stage B's matching head resolves cross-window splits with an Edge Precision of 79.79% (2567/3217) and an Edge Recall of 70.96% (2883/4063). Each true entity fragments into only 1.45 components on average, demonstrating that transitive matching over compressed quotient nodes can successfully reconstruct long-range entities.

8.3 Resource Footprint and Efficiency Analysis

By locking the primary encoder backbones and routing cross-window connections through the quotient graph, the model significantly reduces computational overhead.

As itemized in Table 4, trained parameters account for only 2.38% (16.8M) of the total parameter budget. In inference, this architectural factorization allows the model to achieve a wall-clock throughput of 19.4 ms / 1k tokens on consumer laptop hardware (a single RTX 4060 Laptop GPU). Crucially, the windowed compression

Component	Parameter Count	Status
Stage A (AntecedentScorer)	8.43M	Trained
Stage B (MentionMatcher)	8.40M	Trained
RoBERTa-large	355.00M	Frozen
BGE-large	335.00M	Frozen
Total Parameters	**706.83M**	(2.38% Trained)

Table 4: System parameter budget distribution.

layers prevent the system from materializing the full mention-pair score matrix, lowering peak inference VRAM requirements by **3.1**× compared to full-document all-pairs execution (159 MB vs 488 MB).

9 Limitations and Future Work

Every result in this paper leaves both underlying representation encoders frozen. Consequently, the contextual channel is entirely dependent on general-purpose RoBERTa-large activations with zero coreference-specific fine-tuning. The PRON-only floor (0.6640) highlights this boundary: pronouns carry low semantic signal, forcing the matching head to rely on contextual binding spaces that naturally decay across extreme window gaps. Furthermore, this work assumes gold mention boundaries to isolate coreference resolution logic from mention detection errors. Scaling this quotient-graph architecture to consume predicted mentions from an open span detector remains an immediate avenue for future research.

10 Conclusion

This work demonstrates that global entity resolution can be recovered from local inference plus quotient-graph reconstruction. More importantly, the results provide evidence for a broader computational hypothesis: many language tasks may be decomposable into low-redundancy information channels whose interactions can be modeled independently and recombined through lightweight composition layers. Under such a decomposition, expensive global attention need not operate over the original token space, but only over a compressed latent state space induced by the task structure.

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